



Assessment Cover Sheet

Assessment Title

Designing a Speed Controller for a DC Motor

Assessment Type

Uncontrolled

Individual

Not must-pass

Due Date

December 18, 2024

Course Code

EN7230T

Course Title

Instrumentation and Automatic Control

Internal Moderator's Name

Dr. Saam Najat

External Examiner's Name

Instructions:

1. This cover sheet must be completed (section in red below) and attached to your assessment before submission in hard copy/soft copy.
2. The time allowed for this assessment is till December 18, 2024 @ 11:59 PM.
3. This assessment carries 60 marks distributed to a total of 3 tasks assessing CILO 1 - CILO 4.
4. The **use of generative AI tools is strictly prohibited.**
5. References consulted (if any) must be properly acknowledged and cited.
6. The assessment has a total of 9 pages.

Learner ID

Date Submitted

Learner Name

Programme Code

ENG

Programme Title

Bachelor of Engineering Technology (Electrical)

Lecturer's Name

By submitting this assessment for marking, I affirm that this assessment is my own work.

Learner Signature

Do not write beyond this line. For assessor use only.

Assessor's Name

Marking Date

Marks Obtained

Comments:

Instructions

- 1) Please ensure that you submit your report by the specified due date through the Turnitin link provided on Moodle.
- 2) The date of your upload on Moodle will be considered as the official "Date of Submission" for calculating any late submissions.
- 3) Adhering to the late submission rule is crucial as it will be applied in case of any tardy submissions.
- 4) Kindly note that the deadline for this project is **Wednesday the 18th of December 2024.**

Learning Outcomes

The tasks included in this assignment fulfill the Learning Outcomes of the course as follows:

- 1) Demonstrate advanced knowledge of core principles of control and instrumentation.
- 2) Critically analyze and evaluate transient response characteristics of first and second order systems.
- 3) Apply and demonstrate instrumentation and control strategies for feedforward and feedback control systems.
- 4) Formulate, apply, and present tuning methods for finding suitable PID controller parameters to meet an industry standard requirement for a defined control design problem.

The completion of this project carries a weightage of 60% towards your final grade. The project entails designing a speed controller for a DC motor using **MATLAB/Simulink**. The specific time frame for the project is mentioned above.

Deliverables

By the project deadline, please submit:

- 1) The **MATLAB/Simulink** files used to design your proposed controller.
- 2) A **MATLAB/Simulink** simulation model of the complete control system.
- 3) A project report explaining how you determined the controller design and why you made those particular design choices.

The report should include details of your calculations and reasoning behind the design. Simulation results and analysis can help show the effectiveness of the approach.

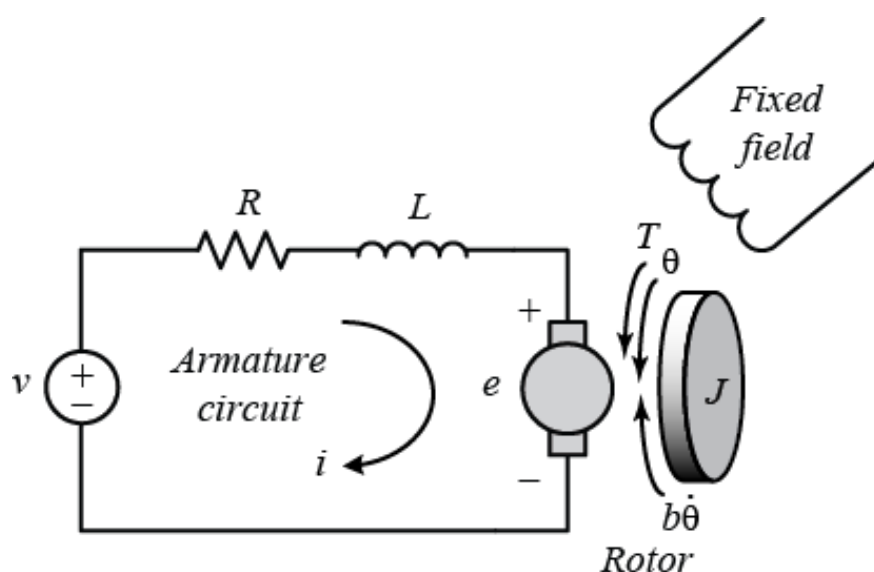
Guidelines for the Project Report

Prepare a comprehensive project report that encompasses the following tasks. Justify your selection of the controller and demonstrate how it meets the criteria for the transient response design specifications. Ensure that the report is clear and transparent by following these guidelines:

- Present your results using appropriate **MATLAB/Simulink** plots and code (**very, very strict rule**).
- Include all m-files (**editable text, no pictures**) and Simulink block diagrams in the appendix section of your report (**not-so-strict rule**).
- Label the axes of each graph, including a figure legend when necessary. Specify the units in the label itself, such as "*speed (m/s)*" or "*time in milliseconds [ms]*," to provide clarity about the measured quantities (**very, very strict rule**).
- To avoid sending multiple figures, consider using the **subplot()** function to group related figures (**not-so-strict rule**).
- Add comments within each m-file to describe the purpose of the **MATLAB** code or command (**very, very strict rule**).
- Give each graph a descriptive title and ensure that all axes are labeled and scaled accurately (**very, very strict rule**).
- If a plot includes multiple lines, include a legend that explains each curve (**very, very strict rule**).
- Do not include **Simulink** 'Scope' images in the report, as they lack proper labeling (**very, very strict rule**).
- When creating **Simulink** block diagrams, minimize overlapping and crossing lines as much as possible. Rearrange icons to establish a clear left-to-right path (**very, very strict rule**).
- Number your plots and graphs sequentially, such as Fig. 1, Fig. 2, and so on (**very, very strict rule**).

DC Motor

The DC motor is a widely used actuator in control systems. It is capable of producing rotary motion directly and, when combined with wheels, drums, or cables, can also generate translational motion. The electric equivalent circuit of the motor's armature and the free-body diagram of the rotor are depicted in the accompanying figure.



In this specific scenario, we will consider the voltage source (V) applied to the motor's armature as the input to the system, while the rotational speed of the shaft ($\dot{\theta}$) will serve as the output. Both the rotor and the shaft are assumed to possess rigid characteristics. Additionally, we adopt a viscous friction model, meaning that the friction torque is proportionate to the angular velocity of the shaft.

The physical parameters for our particular example are as follows:

Parameter	Description	Value
J	moment of inertia of the rotor	$0.05 \text{ kg}\cdot\text{m}^2$
b	motor viscous friction constant	$0.1 \text{ N}\cdot\text{m}\cdot\text{s}$
K_e	electromotive force constant	$0.05 \frac{\text{V}}{\text{rad}\cdot\text{s}}$
K_t	motor torque constant	$0.05 \frac{\text{V}}{\text{rad}\cdot\text{s}}$
R	electric resistance	0.5Ω
L	electric inductance	0.8 H

System's Model Derivation

Typically, the torque produced by a DC motor is directly related to both the armature current and the strength of the magnetic field. However, in this particular case, we will assume that the magnetic field remains constant, resulting in the motor torque being solely proportional to the armature current i . This relationship is expressed by the equation below, where the proportionality factor is denoted as K_t . This configuration is commonly referred to as an *armature-controlled motor*:

$$T = K_t i \quad (1)$$

The back electromotive force (emf), represented as e , is directly proportional to the angular velocity of the shaft. This relationship is governed by a constant factor K_e as follows:

$$e = K_e \dot{\theta} \quad (2)$$

In the International System of Units (SI), the constants for motor torque and back electromotive force (emf) are equivalent, meaning that K_t is equal to K_e . For simplicity, we will denote this common constant as K to represent both the motor torque constant and the back emf constant.

Using the information provided in the preceding figure, we can derive the following governing equations by applying Newton's second law and Kirchhoff's voltage law:

$$J\ddot{\theta} + b\dot{\theta} = Ki \quad (3)$$

$$L \frac{di}{dt} + Ri = V - K\dot{\theta} \quad (4)$$

Transfer Function

By employing the Laplace transform and assuming zero initial condition, we can represent the aforementioned modeling equations, given in (3)-(4), in terms of the Laplace variable s :

$$s(Js + b)\Theta(s) = KI(s) \quad (5)$$

$$(Ls + R)I(s) = V(s) - Ks\Theta(s) \quad (6)$$

By eliminating $I(s)$ between the two equations mentioned in (5)-(6), we obtain the following open-loop transfer function:

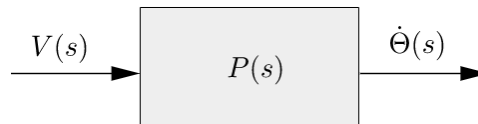
$$P(s) = \frac{\dot{\theta}(s)}{V(s)} = \frac{K}{(Js + b)(Ls + R) + K^2} \quad (7)$$

In this transfer function, the rotational speed $\dot{\theta}(s)$ is regarded as the output, while the armature voltage $V(s)$ is regarded as the input.

Tasks

Part 1: Open-Loop System [15 marks]

In open-loop systems, there is no sensor to measure the output and compare it with the user-defined set-point via a feedback stream, and thus there is no compensator for the mismatch between input and output. This system can be illustrated as follows:

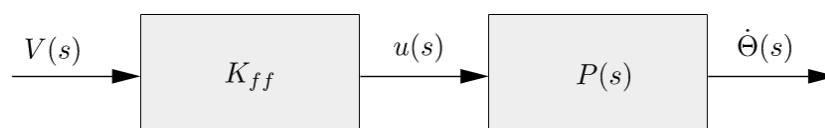


- 1.1 Substitute the numeric values of the system parameters in $P(s)$ and re-express it again. [5 marks]
 - 1.2 Using **MATLAB**, code the above transfer function and find the plot of its step response of 12 volts; i.e. $V(s) = \frac{12}{s}$. [5 marks]
 - 1.3 From these plots, calculate the difference between the input signal and the output signal. Will the output reach the input, or will it have a steady-state error? [5 marks]
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Part 2: Feedforward Control System [22.5 marks]

One of the main objectives of embedding a control system is to ensure that the output signal, represented by $\dot{\Theta}(s)$, closely resembles the input signal, represented by $V(s)$.

If we assume zero external disturbances and a perfect match between the model and the actual system, we can incorporate the following feedforward controller to achieve the objective.

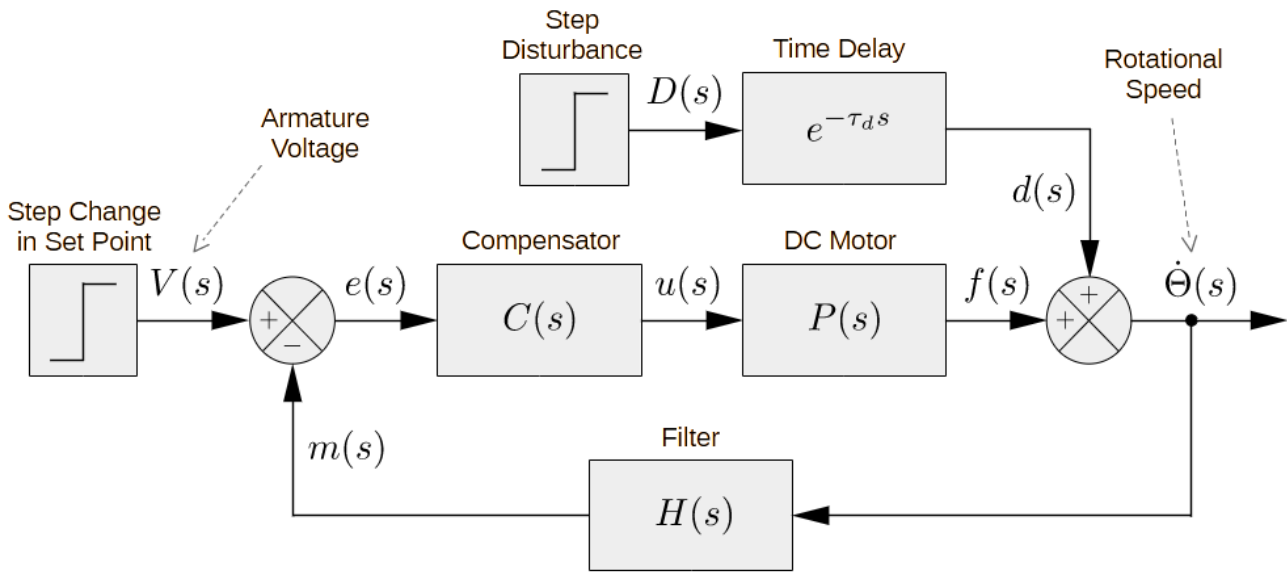


- 2.1 Using the trial-and-error technique, find the value of the feedforward gain (K_{ff}) that satisfies the condition $\dot{\Theta}(s) = V(s)$. [7.5 marks]
 - 2.2 Now, find the inverse of the final value of $P(s)$; i.e., calculate $P(s)^{-1}$ when $s = 0$. [7.5 marks]
 - 2.3 Compare K_{ff} with $P(0)^{-1}$. Are they the same? Why? [7.5 marks]
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Part 3: Feedback Control System [62.5 marks]

The main inherent weakness of the feedforward control scheme is its inability to mitigate uncertainty, such as unknown or unconsidered disturbances and/or imperfections in the model. Also, there is no control on the output performance (such as overshoot, undershoot, raise time, settling time, and offset or steady-state error). To achieve these criteria, a feedback control scheme is suggested.

Assume the following block diagram represents the prospective feedback control scheme where a filter is placed in series with the sensor and the process is affected by an external disturbance:



The disturbance signal is modeled as a step function of magnitude 0.5 and delayed by a time constant of $\tau_d = 0.1$. Thus, the step disturbance in Laplace domain is:

$$D(s) = \mathcal{L}\{0.2\} = \frac{0.5}{s} \quad (8)$$

and the time delay can be expressed as a rational function using the 2nd order Padé approximation:

$$e^{-0.1s} \approx \frac{s^2 - 60s + 1200}{s^2 + 60s + 1200} \quad (9)$$

The measurement unit's filter behaves according to the following function:

$$h(t) = e^{-t} \quad (10)$$

Finally, the compensator is a PID controller whose general expression in Laplace transform is:

$$C(s) = K_p + \frac{K_i}{s} + K_d s \quad (11)$$

- 3.1 Using **Simulink**, model this closed-loop system and connect the input and output signals to a two-channel scope. [10 marks]
- 3.2 Using **MATLAB** or **by hand**, find the overall transfer function between $V(s)$ and $\dot{\Theta}(s)$ and the overall transfer function between $D(s)$ and $\dot{\Theta}(s)$. [10 marks]
- 3.3 Apply the online Ziegler-Nichols (Z-N) tuning method to find the gains (K_p , K_i , and K_d) for P-Controller, PI-Controller, and PID-Controller. [10 marks]
- 3.4 Apply the online Tyreus-Luyben (T-L) tuning method to find the gains (K_p , K_i , and K_d) for PI-Controller and PID-Controller. [10 marks]
- 3.5 Check these tuned PID parameters in your Simulink model and compare between the two tuning methods for PI- and PID-Controller in terms of maximum overshoot, settling time, and steady-state error. Which one has a better result? Why is it better in your opinion? [10 marks]
- 3.6 Use the *optimal PID tuner* (the ready-made **MATLAB** code is attached) to find much better results with a maximum overshoot of 5% or less. [12.5 marks]

Note: there are 3 parts, each task of part 1 weighs 5 marks, each task of part 2 weighs 7.5 marks, and each task of part 3 (except the last task) weighs 10 marks, and the last task of part 3 weighs 12.5 marks.

Plagiarism & Collusion

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